

Environmental Science and Sustainable Engineering

Alternative Assessment Method:
Individual Technical Report

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Green Building Energy Management Portal (GBEMP)

**An AI and IoT-Based Approach for
Improving Building Energy Efficiency**

ABSTRACT

Buildings use a large amount of electricity for air-conditioning, lighting, computers, elevators and other equipment. A major reason for unnecessary energy consumption is that these systems are often used even when they are not needed. For example, air-conditioning may continue running in an empty classroom or lights may remain switched on after working hours.

To address this problem, we propose a **Green Building Energy Management Portal (GBEMP)**. It is an IoT and AI-based web portal that monitors building energy use and helps identify where electricity is being wasted.

The main idea behind GBEMP is to create a **Digital Energy Twin** of the building. It uses data from smart meters, occupancy sensors and environmental sensors to understand the building's normal energy pattern. AI then compares actual energy use with the expected pattern and identifies unusual consumption.

The most important individual contribution is the **AI Energy-Waste Detection Module**. It detects abnormal energy usage, identifies possible reasons for the wastage and provides specific energy-saving recommendations.

Unlike a normal monitoring dashboard, GBEMP follows a complete cycle:

Detect -> Diagnose -> Recommend -> Act -> Verify

This makes the proposed system more useful for real-world buildings such as colleges, offices, hospitals and commercial buildings.

INTRODUCTION

Energy efficiency is becoming increasingly important as buildings consume significant amounts of electricity and contribute to environmental pollution and greenhouse-gas emissions. According to UNEP, the buildings and construction sector is responsible for a significant share of global energy use and CO₂ emissions.

Energy is often wasted because of poor HVAC management, unnecessary lighting, equipment running during non-working hours and lack of real-time information.

Traditional electricity bills only tell us how much energy was consumed. They do not clearly tell us where the energy was wasted or why it happened.

GBEMP attempts to solve this problem by combining sensors, AI and a web-based management system.

PROBLEM STATEMENT

The selected environmental problem is **energy-inefficient buildings**.

Common causes include:

- Air conditioning in unoccupied rooms
- Lights being left on unnecessarily
- Poor temperature control
- Equipment operating after working hours
- High standby power consumption
- Lack of real-time monitoring
- Difficulty identifying the cause of high consumption

Most basic energy-monitoring systems show graphs and electricity readings. However, the building manager still has to manually identify the reason for high consumption.

For example:

Traditional system: "Energy consumption increased by 20%."

GBEMP: "Energy consumption in Room 204 is 20% above its normal level while occupancy is low. HVAC operation is a possible cause. Review the HVAC schedule."

Therefore, GBEMP focuses on understanding energy consumption instead of simply displaying it.

SUSTAINABILITY ANALYSIS

The proposed system supports four important SDGs.

SDG 7 – Affordable and Clean Energy:

The portal encourages efficient use of electricity and supports better energy management.

SDG 11 – Sustainable Cities and Communities:

Energy-efficient buildings contribute to more sustainable cities.

SDG 12 – Responsible Consumption and Production:

The system encourages responsible use of electricity by making consumption visible to users.

SDG 13 – Climate Action:

Reducing unnecessary electricity consumption can reduce associated greenhouse-gas emissions.

The United Nations identifies energy efficiency and sustainable resource use as important elements of these goals.

ISO 14001:

ISO 14001 provides a framework for environmental management and continual improvement.

GBEMP can support this by providing:

- Continuous energy monitoring
- Environmental performance data
- Energy-reduction targets
- Automated reports
- Corrective-action information
- Performance comparison over time

The portal does **not itself provide ISO certification**, but it can support an organization's environmental management system.

LCA and EIA:

Life Cycle Assessment (LCA) can be used to study environmental impacts throughout a building's life cycle. ISO 14040 provides the framework for LCA .

GBEMP mainly contributes to the operational stage, where energy consumption occurs continuously.

A full Environmental Impact Assessment is generally more relevant to physical development projects than to the software portal itself. However, GBEMP data can support environmental decision-making during building construction, renovation and operation.

PROPOSED SOLUTION

Green Building Energy Management Portal (GBEMP)

GBEMP is a web-based system that combines:

IoT + AI + Digital Energy Twin + Web Dashboard

The system collects data from the building, creates a digital representation of its energy behaviour, identifies inefficient patterns and recommends or performs corrective actions.

Main Components:

1. IoT Energy Monitoring Hub

Collects real-time electricity data from smart meters and connected equipment.

2. Occupancy Intelligence Module

Checks whether rooms or zones are occupied and compares occupancy with energy consumption.

3. AI Energy-Waste Detection Module

Identifies unusual energy consumption and possible wastage.

4. Digital Energy Twin

Creates a virtual representation of the building's energy behaviour.

5. Energy Forecast Engine

Predicts future energy consumption and possible peak-demand periods.

6. Smart HVAC and Lighting Control

Can optimize HVAC and lighting according to occupancy and building conditions where compatible control systems are available.

7. Green Action Recommender

Provides specific recommendations instead of general energy-saving advice.

8. Carbon and Cost Dashboard

Shows energy consumption, estimated electricity cost and associated carbon emissions.

9. Building Sustainability Score

Gives each building or zone a dynamic Green Efficiency Score.

10. Admin and Alert System

Alerts building managers about unusual energy consumption, possible equipment problems and high-demand periods.

The key individual contribution to the project is the development of the:

AI Energy-Waste Detection Module.

This module is responsible for analysing building energy and occupancy data to identify unusual consumption and potential energy wastage.

Its main functions are:

- Collect energy-consumption data.
- Collect occupancy information.
- Establish normal energy-consumption patterns.
- Compare actual consumption with expected consumption.
- Identify abnormal energy behaviour.
- Determine possible causes of wastage.
- Estimate potential energy loss.
- Generate personalized recommendations.
- Assign priority to detected problems.
- Track whether the recommended action actually reduced energy consumption.

FEASIBILITY ANALYSIS

Technical Feasibility:

The proposed system is technically feasible because its major components are already available.

Smart meters and IoT sensors can collect building data, while existing web and AI technologies can process and display the information.

Bureau of Energy Efficiency (BEE) also identifies areas such as HVAC, lighting, electrical systems and renewable energy as important parts of building energy efficiency.

Economic Feasibility:

A small prototype can be developed at relatively low cost using:

- Affordable IoT controllers
- Basic sensors
- Open-source software
- Free or low-cost databases/cloud services

For larger buildings, installation costs will increase because more sensors and meters are required.

However, the system is designed to identify energy-saving opportunities that can reduce long-term operating costs.

Actual savings should be measured after implementation.

Operational Feasibility:

Different users can use different parts of the system.

Building Manager:

Views overall energy performance and reports.

Facility Engineer:

Investigates HVAC, lighting and equipment problems.

Administrator:

Sets energy targets and monitors sustainability scores.

Occupants:

Receive energy-saving information and recommendations.

Scalability:

The system can grow from:

One Room -> One Floor -> One Building -> Multiple Buildings -> Smart Campus

This makes it suitable for schools, colleges, offices, hospitals and commercial buildings.

EXPECTED ENVIRONMENTAL BENEFITS

1. Reduced Electricity Consumption

The system identifies unnecessary energy use and provides targeted actions.

2. Reduced Carbon Emissions

Reduced electricity consumption can result in lower associated emissions.

The portal can estimate:

$$\text{CO}_2 = \text{Energy Consumption} \times \text{Emission Factor}$$

The actual emission factor should be selected from an appropriate current source when the system is deployed.

3. Reduced Peak Demand

Energy forecasting can help managers avoid unnecessary high loads during peak periods.

4. Better HVAC and Lighting Management

Occupancy-based information can reduce unnecessary operation of systems.

5. Improved Environmental Awareness

Users can clearly see the connection between energy, cost and carbon emissions.

JUSTIFICATION OF THE PROPOSED SOLUTION

Justification of the Proposed Solution

The proposed Green Building Energy Management Portal (GBEMP) is justified because conventional energy monitoring mainly shows how much electricity a building consumes but does not clearly explain where, why, or when energy is being wasted. GBEMP combines IoT sensors, AI analysis, and a Digital Energy Twin to identify abnormal consumption by comparing actual energy use with expected usage based on occupancy, time, temperature, and past data. It then suggests specific actions, such as adjusting HVAC operation or switching off unnecessary lighting.

The solution is practical because it can use existing smart meters and low-cost IoT sensors, while the software can be developed using commonly available technologies. It also supports real-time monitoring, prediction, automated control, and verification of savings, making it more useful than a basic energy dashboard. By reducing unnecessary electricity consumption, the portal can lower energy costs, carbon emissions, and peak demand, while improving the building's overall sustainability.

Most importantly, the AI Waste Detection Module makes the proposed system unique by turning raw energy data into actionable information rather than simply displaying consumption figures.

Limitations:

Some limitations need to be considered:

- Sensors must provide accurate data.
- AI requires sufficient historical data.
- Large buildings may require significant initial investment.
- Internet connectivity is needed for continuous monitoring.
- AI recommendations depend on data quality.
- Some recommended actions require human approval.
- Occupancy-related data must be handled responsibly.

Future Scope:

In the future, the portal can be expanded with advanced AI and a 3D Digital Energy Twin to simulate different energy-saving strategies before applying them. It can also support predictive maintenance, solar energy integration, mobile alerts, and automatic control of HVAC and lighting. With more buildings connected, the system could develop into a smart campus or city-wide energy management platform, helping organizations reduce energy costs, carbon emissions, and wastage.

CONCLUSION

Energy-inefficient buildings waste electricity through unnecessary HVAC operation, lighting, equipment use and poor energy management. Traditional monitoring systems can show consumption but often do not explain why the consumption is high.

The proposed Green Building Energy Management Portal (GBEMP) addresses this problem using IoT, AI and Digital Energy Twin technology.

The main innovation is the AI Energy-Waste Detection Module, which is the key individual contribution. It analyses energy and occupancy data, identifies unusual consumption, determines possible causes and provides targeted recommendations.

The system creates a continuous cycle of detection, diagnosis, recommendation, action and verification.

This makes the proposed portal more than an energy dashboard. It becomes an intelligent decision-support system that can help buildings reduce energy wastage, operating costs and associated environmental impacts.

With further development, GBEMP can be expanded from a single building to an entire smart campus or group of buildings, making it a practical technology-based approach to sustainable building management.

REFERENCES

- [1] United Nations Environment Programme, *Global Status Report for Buildings and Construction 2025–2026*, UNEP, 2026. [UNEP – Global Status Report for Buildings and Construction](#)
- [2] United Nations, *Sustainable Development Goals*, United Nations Department of Economic and Social Affairs. [UN Sustainable Development Goals](#)
- [3] International Organization for Standardization, *ISO 14001:2026 – Environmental management systems — Requirements with guidance for use*, ISO, 2026. [ISO 14001](#)
- [4] International Organization for Standardization, *ISO 14040:2006 – Environmental management — Life cycle assessment — Principles and framework*, ISO, 2006. [ISO 14040](#)
- [5] Bureau of Energy Efficiency, Government of India, *Energy Conservation Building Code (ECBC)*, BEE. [Bureau of Energy Efficiency](#)